Correction budget measured on sub-areas of the full-sky Uchuu lightcone (SFD E(B-V), $|b| > 25^\circ$, 23794 deg^2) and extrapolated to large surveys

Fit precision of the systematic amplitude

$\sigma_\varepsilon = C_\varepsilon\,(A/100\,\mathrm{deg}^2)^{-\alpha}$, $\alpha = 0.27, 0.29, 0.32$

$\sigma_\varepsilon^2 = f^2(1/\bar{N} + e^{\sigma^2} - 1)\,A_\mathrm{pix}/A$, $f = 1.62\,(A/100)^{0.19}$

Bias error from the sub-area scatter

$\sigma_b = C_b\,(A/100\,\mathrm{deg}^2)^{-\beta}$, $\beta = 0.40, 0.42, 0.45$

$b = \sqrt{e^{\sigma^2} - 1}\,/\,\sigma_{m,\,\mathrm{pix}}$

Safe correction window

$\varepsilon_\mathrm{tol} = \sigma_{m,\,\mathrm{pix}}\sqrt{(b+\sigma_b)^2 - b^2}$

$|k-1| < \varepsilon_\mathrm{tol}/\varepsilon$, reliable while $\sigma_\varepsilon < \varepsilon_\mathrm{tol}$

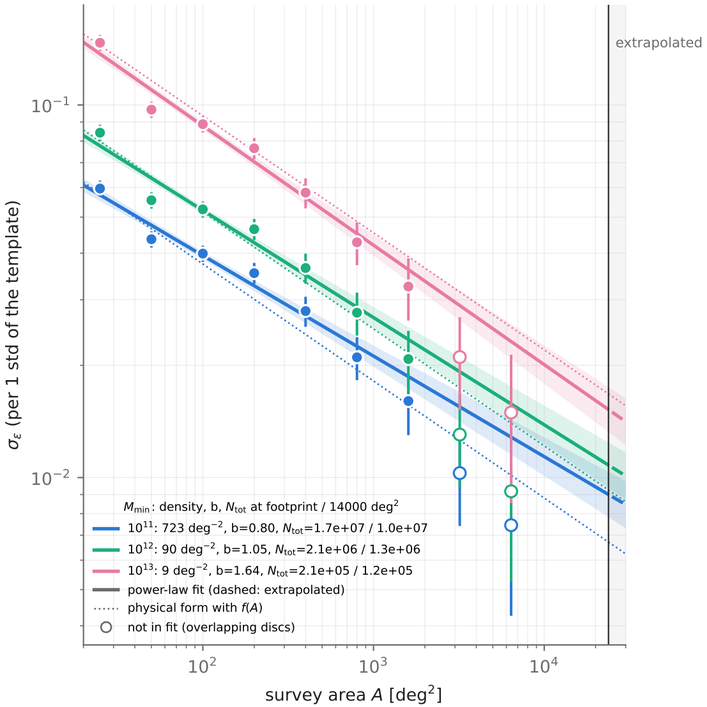
t normalised over each (sub-)survey

Mis-correction offset, rule of thumb

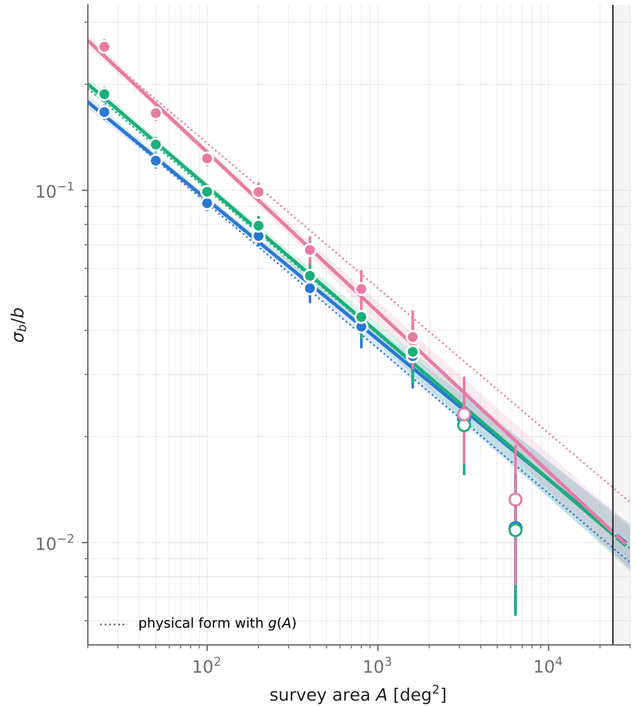
$b'^2 = b^2 + (1-k)^2\varepsilon^2/\sigma^2_{m,\,\mathrm{pix}}$, offset $(b'-b)/\sigma_b$

$\sigma_b/b = g\,\frac{\sqrt{2}}{2}\,\frac{1/\bar{N}+v}{v}\sqrt{A_\mathrm{pix}/A}$, $v = b^2\sigma^2_{m,\,\mathrm{pix}}$, $g = 1.38\,(A/100)^{0.09}$

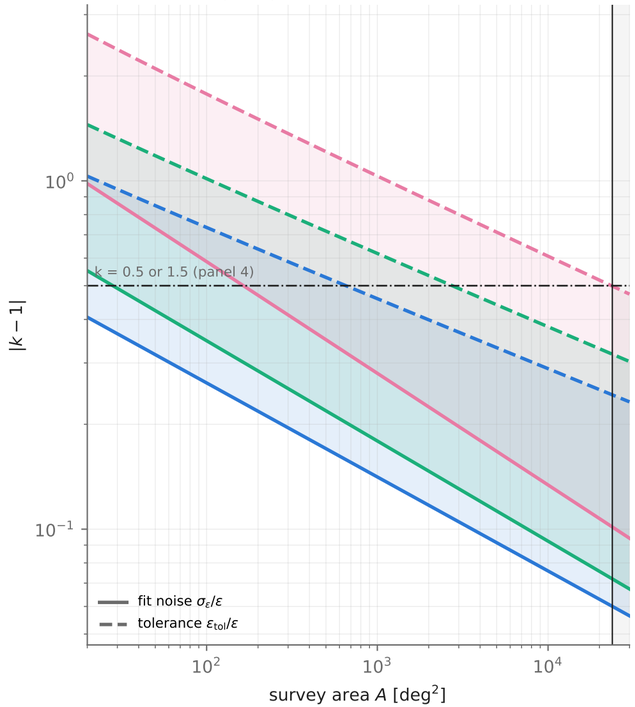
1. Fit precision σ_ε of the amplitude



2. Relative bias error σ_b/b



3. Safe window $|k-1|$ (shaded: correctable)



4. Bias offset of a 50% mis-correction (k = 1 gives 0)

